SREEKALA K R

SOFTWARE ENGINEER

sreekalakr22@gmail.com | +91 6238415930 | Ernakulam, Kerala, India | GitHub | LinkedIn

PROFESSIONAL SUMMARY

Results-driven Software Engineer with nearly 3 years of experience designing, developing, and delivering responsive, scalable web applications. Proficient in Angular, React, TypeScript, and Node.js, with a strong foundation in RESTful API integration, microservices architecture, authentication systems, and performance optimization. Demonstrated success in IoT-enabled maritime platforms, real-time data handling, and full-stack feature delivery. Adept at collaborating in Agile/Scrum teams, mentoring peers, and solving complex technical challenges across diverse environments.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, Python
Frontend: Angular (14+), React, Next.js, HTML5, CSS3, SCSS, Bootstrap, Tailwind CSS
Backend: Node.js, Express.js, NestJS, Django
Databases & Storage: MySQL, PostgreSQL, Cassandra (DataStax Astra), MongoDB, Redis, SQLite
ORMs & ODMs: Sequelize, Prisma, TypeORM, Mongoose
Cloud & Dev Tools: AWS (S3, DynamoDB, CloudWatch), Git, GitHub, Postman, VS Code, Jira, Linear, Figma
Authentication & Authorization: Auth0, JWT, OAuth 2.0, SSO, RBAC
AI Tools & Platforms: ChatGPT, Cursor AI, Claude, GitHub Copilot, Gemini
Architecture & Practices: Microservices Architecture, RESTful APIs, Performance Optimization, Agile/Scrum, Code Review
Operating Systems: Windows, Linux

PROFESSIONAL EXPERIENCE

Software Engineer | **ZeroNorth**, Chennai, India *November 2023 – February 2026*

Project: SMARTShip – *IoT-enabled vessel management platform for real-time monitoring of vessel performance, machinery health, safety, and compliance.*

- Developed reusable Angular (14+) components using TypeScript, improving maintainability and reducing duplicate code across the platform.
- Revamped and reskinned the full application UI using HTML5, CSS3, and SCSS, improving navigation flow, usability, and cross-device responsiveness across desktop and mobile viewports.
- Leveraged RxJS for reactive programming and efficient asynchronous data handling across complex real-time vessel monitoring workflows, minimizing latency and UI blocking.
- Applied Angular performance best practices, including lazy loading, to reduce bundle size and improve rendering speed on low-bandwidth maritime networks.
- Migrated client-side storage from LocalStorage to IndexedDB to efficiently manage large real-time vessel sensor datasets, improving data reliability and offline resilience.
- Integrated and consumed RESTful APIs built on Node.js and NestJS within a microservices architecture to support real-time vessel monitoring, reporting, and compliance workflows.
- Utilized Redis for session caching and high-frequency vessel telemetry data, reducing backend query load and improving API response times across critical dashboard endpoints.
- Integrated DataStax Astra (Cassandra) to store vessel telemetry data for historical analysis and fleet reporting.
- Collaborated with backend teams to design and optimize PostgreSQL schemas using TypeORM and MongoDB collections using Mongoose for high-frequency IoT data ingestion from multiple vessel sources.
- Integrated Auth0 implementing OAuth 2.0 and JWT-based token flows along with Single Sign-On (SSO), delivering secure and streamlined user access management across enterprise clients.
- Designed and enforced role-based access control (RBAC) across the application, ensuring fine-grained permission management aligned with enterprise security requirements.

- Automated voyage-related workflows including configuration, scheduling, and alerting using NestJS backend modules, significantly reducing manual operational effort and improving fleet efficiency.
- Architected automated email notification workflows for user onboarding, password recovery, duplicate account handling, and team invitations using Node.js mail services.
- Implemented Mixpanel event tracking to capture user behavior, feature adoption, and engagement metrics, providing data-driven insights to guide product roadmap decisions.
- Monitored application reliability and performance using AWS CloudWatch dashboards and custom alerts, enabling proactive issue detection and maintaining high system availability in production.
- Stored and retrieved vessel data files using AWS S3, ensuring reliable and scalable object storage for reporting modules.
- Utilized AWS DynamoDB for real-time storage and low-latency retrieval of live vessel operational data, enabling fast dashboard updates and critical monitoring workflows.
- Upgraded Node.js and Angular versions across Linux-based development and staging environments, resolving dependency conflicts and enhancing long-term security compliance and stability.
- Used Postman to develop, test, and document RESTful API contracts; tracked sprint tasks and releases using Jira and Linear within Agile/Scrum delivery cycles.
- Developed supplementary React-based interfaces for internal reporting modules, leveraging component-driven architecture for rapid UI iteration.
- Leveraged AI-assisted development tools including GitHub Copilot, Cursor AI, ChatGPT, and Gemini to accelerate feature development, code reviews, and technical documentation.
- Participated actively in code reviews and mentored junior developers, promoting best practices in Angular, RxJS, TypeScript, NestJS, and clean code principles.
- Collaborated cross-functionally with Product, QA, and DevOps teams in Agile sprints to deliver client-focused features on schedule and aligned with business requirements.

Software Engineer Intern | **Alpha Ori India Private Limited**, Chennai, India *April 2023 – October 2023*

Project: Smart Voyager – *Voyage planning and decision-support platform focused on route optimization, fuel efficiency, and operational safety.*

- Developed and maintained frontend features using Angular and TypeScript within a scalable, component-based architecture, building responsive interfaces that delivered a consistent user experience across all screen sizes.
- Integrated RESTful APIs using Node.js and Express.js to support dynamic data retrieval, real-time alerting, and dashboard updates for voyage planning workflows.
- Implemented interactive charts and data visualization modules for real-time and historical voyage data reporting, enabling fleet operators to monitor fuel consumption and route efficiency.
- Assisted in building backend services using Express.js, writing SQL queries against MySQL databases and working with Mongoose for MongoDB-based document storage.
- Implemented JWT-based authentication flows on the frontend to secure protected routes and maintain session state across the single-page application.
- Participated in UI/UX review sessions referencing Figma mockups and validated API endpoints using Postman to ensure implementation matched design and backend specifications.
- Managed feature branches and collaborative development workflows using Git and GitHub; tracked sprint tasks and bugs using Jira in an Agile team environment.
- Collaborated in Agile development cycles contributing to debugging, testing, and feature delivery under senior developer guidance, gaining domain knowledge in maritime fuel management and route optimization.

EDUCATION

B.Tech in Computer Science and Engineering

Mar Athanasius College of Engineering, Kothamangalam | 2020 – 2023

Diploma in Computer Engineering

Govt. Polytechnic College, Kothamangalam | 2017 – 2020

ADDITIONAL INFORMATION

Languages Known: English, Malayalam; Basic proficiency in Tamil and Hindi